

Phylogenetic Analysis Kinesin-1 (KIF5B) and Kinesin-14 (KIFC1)

GitHub Link: <https://github.com/Rameensajjad/Phylogenetic-Analysis>

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1. Executive Summary

This report presents a comprehensive maximum-likelihood (ML) phylogenetic analysis of kinesin motor proteins, focusing on two functionally and structurally divergent subfamilies: Kinesin-1, represented by human KIF5B (NP_004512), and Kinesin-14, represented by human KIFC1/HSET (NP_002254). These two proteins were selected because they exemplify the two extremes of kinesin motor domain architecture — N-terminal plus-end directed motors versus C-terminal minus-end directed motors — making them ideal anchors for probing the deep evolutionary history of the kinesin superfamily.

Using a fully automated bioinformatics pipeline executed on Google Collaboratory, homologous sequences were retrieved from the NCBI Protein Database via BLASTp searches against the Protein Data Bank (PDB). Retrieved sequences were aligned using MAFFT's L-INS-I algorithm, trimmed to the conserved motor domain core, and subjected to ML tree inference under the LG+I+G4 substitution model as selected by ModelTest-NG across three information criteria. Node confidence was evaluated through 100 non-parametric bootstrap replicates, and ancestral amino acid states were reconstructed at all internal nodes using RAxML-NG marginal inference.

The analysis encompassed 31 unique sequences reduced to 22 non-redundant representatives after deduplication. The ML tree (final lnL = -4101.567) resolved two strongly supported monophyletic clades — one for each kinesin subfamily — separated by a pairwise sequence distance of approximately 0.655, corresponding to ~34.5% identity at the motor domain. Bootstrap support exceeded 95% at the node separating the two clades, and 100% at the deepest root split, confirming that the divergence between Kinesin-1 and Kinesin-14 is ancient, robust, and biologically meaningful.

Key Finding: Kinesin-1 and Kinesin-14 form fully resolved, well-supported monophyletic clades with ~65% sequence divergence at the motor domain level, consistent with an ancient divergence event predating the split of animals and fungi. The conserved ATP-binding and microtubule-binding interfaces are identifiable in all reconstructed ancestral sequences, suggesting that directional force generation is an ancestrally constrained property of the kinesin fold.

2. Introduction and Biological Background

2.1 The Kinesin Superfamily: An Overview

Kinesins are a vast and evolutionarily ancient superfamily of cytoskeletal motor proteins present in virtually all eukaryotes. They were first identified in the 1980s as the molecular motors responsible for fast anterograde axonal transport in squid giant axons, but have since been recognised as essential players in almost every aspect of eukaryotic cell biology. From the movement of mitochondria and lysosomes to the precise choreography of chromosome segregation during mitosis and meiosis, kinesins are indispensable molecular machines.

The defining feature of all kinesins is the motor domain — a globular catalytic core of approximately 340–380 amino acids that binds both ATP and microtubules, coupling nucleotide hydrolysis to

conformational changes that produce mechanical force and directed movement. Despite the extraordinary diversity of kinesin function, this motor domain is structurally conserved across all 14 recognised kinesin families, sharing the same overall fold: a central eight-stranded beta sheet flanked by three alpha helices on each side, with the nucleotide-binding site at the top and the microtubule-binding interface at the bottom.

The 14 kinesin families (Kinesin-1 through Kinesin-14) differ from one another in: the position of the motor domain within the full-length protein (N-terminal, middle, or C-terminal); the direction of movement along microtubules (towards the plus-end or minus-end); the architecture of the non-motor regions (coiled-coil stalks, cargo-binding tails, regulatory domains); oligomeric state (monomers, dimers, tetramers); and biological function.

2.2 Kinesin-1: The Conventional Kinesin

Kinesin-1, the founding member of the superfamily, is the "conventional kinesin" originally discovered in squid axons. In humans it is encoded by three genes — KIF5A, KIF5B, and KIF5C — which are expressed in different tissues and have distinct cargo preferences. KIF5B is the ubiquitously expressed isoform and the most extensively characterised. As a homodimer, KIF5B walks processively towards microtubule plus-ends, taking ~8 nm steps per ATP hydrolysis cycle, and can travel distances of micrometres without dissociating from the microtubule. Its non-motor regions include a long coiled-coil stalk and a C-terminal tail that interacts with light chain subunits (KLC1/2) and directly with cargo.

Physiologically, KIF5B is responsible for anterograde transport of mitochondria, lysosomes, endosomes, and mRNA granules in neurons and other cell types. It also participates in positioning the mitotic spindle and moving the endoplasmic reticulum towards the cell periphery. Loss of KIF5B function leads to perinuclear clustering of organelles and severe cellular dysfunction.

Structurally, the KIF5B motor domain is one of the best-characterised protein structures in existence, with dozens of crystal structures available in the PDB capturing it in different nucleotide states (apo, ADP-bound, ATP-analogue-bound) and in complex with microtubules (cryo-EM). The motor domain contains the highly conserved Walker A (P-loop, sequence GQTSSGKT), Walker B, switch I, and switch II motifs that coordinate Mg^{2+} -ATP binding and hydrolysis, as well as loop L11, which makes the primary contact with the microtubule surface.

2.3 Kinesin-14: The Reverse Kinesin

Kinesin-14 represents the minus-end directed branch of the kinesin superfamily. Unlike Kinesin-1, where the motor domain is at the N-terminus, Kinesin-14 members have their motor domain at the C-terminus of the protein. This inversion of domain organisation is associated with opposite directionality: Kinesin-14 motors walk towards microtubule minus-ends. The molecular basis for this directional switch lies in subtle but critical differences in the orientation of the neck linker — a short peptide that connects the motor domain to the coiled-coil stalk and whose conformational change during the ATPase cycle determines the direction of movement.

In humans, KIFC1 (also known as HSET — Human homolog of the *Xenopus* XCTK2) is the primary Kinesin-14 family member. It is non-processive as a monomer but can crosslink antiparallel microtubules and slide them apart, making it essential for bipolar spindle assembly, particularly in cells with supernumerary centrosomes. Crucially, cancer cells with centrosome amplification — a common feature of aggressive tumours — depend on KIFC1 for clustering extra centrosomes into pseudo-bipolar spindles, a mechanism of oncogene-induced mitotic rescue. This makes KIFC1 an attractive cancer drug target, since its inhibition selectively kills cells with centrosome amplification while leaving normal cells largely unaffected.

The Kinesin-14 family has a remarkably broad phylogenetic distribution, with well-characterised members in *Drosophila melanogaster* (Ncd — the prototypic Kinesin-14), *Saccharomyces cerevisiae* (Kar3), *Schizosaccharomyces pombe* (Pkl1), *Aspergillus nidulans* (klpA), *Xenopus laevis* (XCTK2), and plants (*Arabidopsis* KCH proteins). This breadth makes Kinesin-14 an ideal family for phylogenetic analysis spanning major eukaryotic lineages.

2.4 The Evolutionary Question

The central evolutionary question addressed by this analysis is: given that Kinesin-1 and Kinesin-14 share the same overall motor fold yet differ so fundamentally in domain organisation, directionality, and function, what is the nature and depth of their evolutionary relationship? Are they monophyletic within a broader kinesin superfamily tree? How conserved are their motor domain sequences across eukaryotes? And can ancestral reconstruction identify the sequence features that were present in their last common ancestor?

3. Materials and Methods

3.1 Computational Environment and Software

All analyses were performed using Python 3.12 on Google Colaboratory (CPU runtime, Intel Xeon @ 2.20 GHz, 12 GB RAM, Ubuntu 22.04). The choice of a cloud-based platform reflects the increasingly practical nature of phylogenomic analyses in accessible environments without dedicated HPC infrastructure.

Software	Version	Role in Pipeline
BioPython	1.87	Sequence retrieval, BLAST XML parsing, alignment I/O, tree visualisation
MAFFT	v7.490 (2021/Oct/30)	Multiple sequence alignment using L-INS-i strategy
RAxML-NG	v1.2.0 (09.05.2023)	ML tree inference, bootstrap analysis, ancestral reconstruction

Software	Version	Role in Pipeline
ModelTest-NG	v0.1.7	Amino acid substitution model selection
matplotlib	system	Phylogenetic tree figure generation
NCBI Entrez API	REST	Remote sequence retrieval from NCBI databases

RAxML-NG and ModelTest-NG were installed as precompiled binaries directly from their GitHub release pages, avoiding dependency on conda or package managers that caused checksum verification failures in the Colab environment. BioPython was installed via pip and MAFFT via apt-get.

3.2 Query Sequence Retrieval

The two query protein sequences were retrieved programmatically from NCBI using BioPython's Entrez.efetch() function with the following parameters: database = "protein", return type = "fasta", and the respective accession numbers. Each record was assigned a descriptive identifier (KIF5B_Hs_Kinesin1 and KIFC1_Hs_Kinesin14), stripped of its original NCBI description line to prevent downstream parsing issues, and saved to individual FASTA files in the working data/ directory. A mandatory 0.5-second inter-request delay was imposed to comply with NCBI's API rate-limiting policies (maximum 3 requests per second without an API key).

The two retrieved sequences confirm expected properties: KIF5B at 963 amino acids includes the N-terminal motor domain, the coiled-coil stalk, and the C-terminal cargo-binding tail, while KIFC1 at 673 amino acids includes the N-terminal stalk and the C-terminal motor domain, consistent with the Kinesin-14 domain architecture.

3.3 Homolog Identification via BLASTp

Homologous sequences were identified through BLASTp (protein-protein BLAST) searches against the Protein Data Bank (PDB) using NCBI's remote BLAST service (NCBIWWW.qblast). The PDB was selected as the target database rather than nr (non-redundant) or UniProtKB to: (i) restrict hits to structurally characterised proteins with known three-dimensional conformations; (ii) avoid the massive sequence redundancy in the nr database that would inflate the dataset with nearly identical sequences; and (iii) ensure biological relevance, as PDB entries are predominantly from well-studied model organisms with robust functional annotations.

Each search requested the top 30 hits (hitlist_size=30) and ran for approximately 60–75 seconds on NCBI's remote servers. BLAST XML output was saved locally and parsed using BioPython's NCBIXML module. For each alignment-HSP pair, the following statistics were extracted and

tabulated: hit identifier (PDB accession), total sequence length, alignment length, expect value (E-value), number of identical positions, percent identity (PID%), and query coverage (Cov%).

The percent identity filter was set at PIDcut = 1.00, accepting all hits regardless of identity level. This liberal filter was intentional — for phylogenetic purposes, even distant homologs at 40–45% identity provide valuable evolutionary signal, and false positives are controlled by the subsequent alignment and tree-building steps.

3.4 Dataset Construction

The top 15 hits from each BLASTp search were individually fetched from NCBI's protein database using Entrez.efetch(), again with a 0.4-second inter-request delay. All 30 fetched sequences were pooled with the two original query sequences and subjected to exact-sequence deduplication: each sequence was converted to uppercase and stored as a dictionary key, with duplicate sequences having their identifiers merged with an underscore separator rather than being silently discarded. This approach preserves identity information for sequences that happen to be identical.

After deduplication, the dataset comprised 31 unique sequences from diverse taxonomic origins including *Homo sapiens*, *Rattus norvegicus*, *Bos taurus*, *Drosophila melanogaster*, *Schizosaccharomyces pombe*, *Saccharomyces cerevisiae*, and *Aspergillus nidulans*, representing the full evolutionary breadth of Kinesin-1 and Kinesin-14 detectable in the PDB.

3.5 Multiple Sequence Alignment with MAFFT

The 31 sequences were aligned using MAFFT v7.490 invoked with the --auto flag. MAFFT's auto-detection algorithm evaluated sequence properties (length, number of sequences, degree of divergence) and selected the **L-INS-i** strategy, described by MAFFT as "Probably most accurate, very slow — iterative refinement method with LOCAL pairwise alignment information." L-INS-i is the recommended strategy for protein families where a conserved domain is embedded within variable flanking regions — precisely the situation with kinesins, where the ~370-residue motor domain is flanked by highly variable stalk and tail regions.

The alignment was performed using the BLOSUM62 substitution matrix, gap opening penalty of −1.53, and gap extension of 0.00. The algorithm first built a UPGMA guide tree, performed progressive alignment, and then iteratively refined the result over 16 cycles until convergence. The final alignment comprised 31 sequences across 1,324 columns, with approximately 60–70% of columns containing at least one gap character due to the length variation between full-length proteins and truncated crystallographic constructs present in the PDB.

3.6 Alignment Trimming

The full-length 1,324-column alignment was trimmed to remove uninformative, gap-rich terminal regions that could introduce noise into tree inference. The trimming strategy was conservative: the alignment was scanned from the left to find the first column containing no gap characters in any of the 31 sequences (position 314), and from the right to find the last fully ungapped column (position

685). The alignment was then sliced to columns 314–685, yielding a trimmed alignment of 371 columns.

This 371-column region corresponds precisely to the conserved kinesin motor domain core, encompassing the beta-sheet framework, the Walker A (P-loop) and Walker B motifs, switch I and switch II loops, and the microtubule-binding loop L11. The trimmed alignment contains the overwhelming majority of phylogenetically informative sites while excluding the highly gapped flanking regions that would otherwise dominate the alignment length without contributing positional information.

Original length: 1,324 columns → Trimmed length: 371 columns (28% of original, but containing the core informative region).

3.7 Deduplication of Trimmed Alignment

After trimming, some sequences that were distinct in the full-length alignment became identical within the 371-column motor domain window — a common occurrence when multiple crystal structures represent the same protein construct crystallised under different conditions, with different ligands, or in different space groups. Nine such duplicate sequences were identified and removed, reducing the dataset from 31 to 22 unique sequences. This cleaned alignment (clear_aligned_trimmed.fasta) was used as the sole input for model selection and tree inference.

3.8 Amino Acid Substitution Model Selection

ModelTest-NG v0.1.7 evaluated candidate amino acid substitution models on the 22-sequence cleaned alignment using maximum likelihood scoring under three information criteria. The top 10 models ranked by AICc are shown below:

Rank	Model	Parameters (K)	lnL	AICc Score	Δ AICc
1	LG+G4	1	-4102.075	8299.151	0.000
2	LG+I+G4	2	-4101.875	8300.751	1.600
3	VT+I+G4	2	-4106.325	8309.651	10.500
4	VT+G4	1	-4107.734	8310.468	11.318
5	LG+G4+F	20	-4082.844	8311.689	12.538
6	LG+I+G4+F	21	-4082.848	8314.697	15.546
7	RTREV+G4+F	20	-4086.382	8318.763	19.612

Rank	Model	Parameters (K)	lnL	AICc Score	Δ AICc
8	RTREV+I+G4+F	21	-4086.235	8321.470	22.319
9	WAG+I+G4	2	-4113.159	8323.318	24.167
10	WAG+G4	1	-4114.278	8323.556	24.405

The **LG substitution matrix** dominated the top rankings across all three criteria, confirming it as the most appropriate empirical amino acid replacement model for this dataset. The LG matrix (Le & Gascuel 2008) was derived from a large set of diverse protein alignments and is widely recognised as the most accurate general-purpose amino acid substitution model available.

LG+I+G4 was selected for all downstream analyses rather than the strictly top-ranked LG+G4, because the addition of a proportion of invariant sites (I) provides a more biologically realistic description of rate heterogeneity: some positions in the kinesin motor domain — particularly the catalytic residues directly involved in ATP coordination (Lys at P-loop position 6, Thr at switch I, Asp at switch II) — are effectively invariant across all known kinesins, while others are highly variable. The Δ AICc penalty of only 1.6 units for the additional invariant-sites parameter is well justified by this biological rationale.

Key model parameter estimates: Gamma shape $\alpha \approx 0.730$ – 1.02 (moderate rate variation), P-inv ≈ 0.076 – 0.179 (7–18% invariant sites), using the LG frequency vector.

3.9 Maximum Likelihood Tree Inference

The ML tree was inferred using RAxML-NG v1.2.0 with a multi-start strategy: 10 random starting trees and 10 parsimony-based starting trees, for a total of 20 independent ML searches. Using multiple starting trees guards against the risk of converging on local rather than global optima in the likelihood landscape — a known issue with complex tree spaces for datasets of this size. The search used SPR (subtree pruning and regrafting) moves with automatic radius detection, fast NR branch length optimisation, and site-repeat acceleration for computational efficiency.

Fixed random seed 12345 was used throughout for full reproducibility. The analysis used 2 PTHREADS with coarse-grained parallelisation (2 workers $\times$ 1 thread each). The 20 ML searches produced trees with highly similar log-likelihoods, with the best and worst trees differing by less than 1.0 lnL units, indicating good convergence. Total inference time was approximately 103 seconds.

RAxML-NG issued a warning that the best ML tree contained 9 near-zero branches, which were collapsed in the "best collapsed" tree (T1.raxml.bestTreeCollapsed) but retained in the standard best tree (T1.raxml.bestTree) used for analysis, as near-zero branches often represent genuine polytomies among highly similar sequences rather than topological errors.

3.10 Bootstrap Analysis

Non-parametric bootstrap resampling was performed using RAXML-NG's --bootstrap mode with 100 replicates. Each bootstrap replicate randomly resamples alignment columns with replacement, infers an ML tree under LG+I+G4, and the resulting 100 bootstrap trees are used to calculate Felsenstein Bootstrap Proportions (FBP) — the percentage of bootstrap replicates that recover each bipartition of the best ML tree. Bootstrap values at each internal node represent the frequency (0–100%) with which that node's branching pattern was recovered across the 100 replicates. The 100 bootstrap analyses completed in approximately 290 seconds and were mapped onto the best ML tree using RAXML-NG's --support mode, producing a tree file (T3.raxml.support) in which internal node labels carry bootstrap percentages.

3.11 Ancestral Sequence Reconstruction

Marginal ancestral amino acid state reconstruction was performed using RAXML-NG's --ancestral mode on the best ML tree. This approach produces, for each of the 21 internal nodes of the 22-taxon tree, a 371×20 probability matrix giving the marginal posterior probability of each amino acid at each position. The most probable amino acid at each position is the MAP (maximum a posteriori) estimate of the ancestral residue. The ASR analysis re-estimated branch lengths under LG+I+G4 (final lnL = -4101.897) and produced two output files: the annotated ancestral tree (ASR.raxml.ancestralTree) and the full probability matrix (ASR.raxml.ancestralProbs). Pattern compression was intentionally disabled to ensure accurate per-site probability computation.

4. Results

4.1 BLASTp Search Performance and Hit Profiles

Both BLASTp queries completed within 60–76 seconds on NCBI's remote servers, each returning the requested 30 hits. The hit profiles for the two queries were qualitatively distinct, reflecting the different evolutionary conservation profiles of Kinesin-1 and Kinesin-14.

For **KIF5B (Kinesin-1)**, the top hits showed an unusually steep gradient: the first 16 hits all had E-values of 0.00 and percent identities of 97.9–100%, reflecting the extreme conservation of the human KIF5B motor domain across the many crystal structures of this protein deposited in the PDB. The gradient then dropped through hits at 85–87% identity (rat and bovine KIF5C), 75.8% (a divergent kinesin), 55.6% (Eg5/KIF11, Kinesin-5), and finally 42–45% for the most distant hits, which included Kinesin-14 members (Ncd, Kar3, Pkl1) cross-detected by the conserved motor fold.

For **KIFC1 (Kinesin-14)**, the first hit was KIFC1 itself at 99.7% identity, followed immediately by a cliff-edge drop to ~41–45% identity for all subsequent hits — consistent with the greater sequence divergence within the Kinesin-14 family and the relatively fewer KIFC1-like structures in the PDB compared to Kinesin-1.

Selected KIF5B (Kinesin-1) BLASTp Hits

PDB Hit	Aln Len	E-value	PID%	Cov%	Organism / Functional Note
pdb 8RHB K	963	0.00	100.0	100.0	Human KIF5B full-length — self-hit
pdb 1MKJ A	349	0.00	100.0	36.2	Human KIF5B motor domain, ADP-bound crystal structure
pdb 1BG2 A	325	0.00	100.0	33.7	Human KIF5B motor domain, apo state
pdb 8IXA S	349	0.00	99.7	36.2	Human KIF5B, near-identical construct
pdb 4HNA K	349	0.00	98.3	36.2	Human KIF5B, inhibitor-bound
pdb 3J8X K	349	0.00	98.9	36.2	Human KIF5B, cryo-EM with microtubule
pdb 3J6H K	342	0.00	86.8	35.3	Rattus norvegicus KIF5C, cryo-EM structure
pdb 3WRD A	334	0.00	87.1	34.5	Rattus norvegicus KIF5C, X-ray
pdb 4UXT C	334	0.00	85.3	34.6	Bovine KIF5C motor domain
pdb 2Y5W A	355	0.00	75.8	36.4	Divergent kinesin motor domain
pdb 2KIN A	238	3.9e-148	85.3	24.6	Rattus norvegicus kinesin — first ever kinesin crystal structure
pdb 1GOJ A	347	8.1e-126	55.6	35.4	Eg5/KIF11 (Kinesin-5) — mitotic motor
pdb 1IA0 K	372	3.9e-84	42.5	35.1	Ncd (<i>Drosophila melanogaster</i> Kinesin-14)
pdb 3B6U A	342	2.6e-80	45.3	33.7	Kar3 (<i>S. cerevisiae</i> Kinesin-14)
pdb 1I5S A	363	3.7e-79	42.1	34.2	<i>S. cerevisiae</i> Kinesin-14, ATP-bound

Selected KIFC1 (Kinesin-14) BLASTp Hits

PDB Hit	Aln Len	E-value	PID%	Cov%	Organism / Functional Note
pdb 5WDH A	357	0.00	99.7	53.0	Human KIFC1 motor domain — self-hit
pdb 2NCD A	433	5.5e-97	41.8	63.3	Ncd (Drosophila melanogaster) — prototypic Kinesin-14
pdb 4GKR A	378	1.2e-95	44.7	54.4	Fungal Kinesin-14 homolog
pdb 5W3D A	421	1.9e-95	42.5	61.5	Kinesin-14, AMPPNP-bound state
pdb 1CZ7 A	413	7.2e-95	42.4	60.3	Ncd in complex with ADP
pdb 3U06 A	421	1.5e-94	42.3	61.5	Kinesin-14, transition state mimic
pdb 8YUE A	421	3.6e-94	42.3	61.5	Recently deposited Kinesin-14 structure
pdb 4ETP A	412	6.4e-94	41.5	59.7	Kinesin-14 motor domain
pdb 3T0Q A	371	1.2e-90	45.8	53.5	Pkl1 (Schizosaccharomyces pombe) — fission yeast Kinesin-14
pdb 3KAR A	367	2.2e-86	43.6	52.5	kfpA (Aspergillus nidulans) — fungal Kinesin-14
pdb 1F9W A	369	5.6e-86	43.1	52.7	Ncd mutant, directionality switch study
pdb 4H1G A	364	1.2e-76	42.3	53.3	Kinesin-14, C-terminal motor domain
pdb 5WDE A	353	3.7e-75	43.3	52.5	KIFC1 related construct, different lattice

PDB Hit	Aln Len	E-value	PID%	Cov%	Organism / Functional Note
pdb 3H4S A	368	1.4e-74	39.7	54.7	Divergent Kinesin-14 homolog
pdb 1SDM A	364	4.1e-71	40.1	53.5	Kar3 (<i>S. cerevisiae</i>), Kinesin-14

4.2 Dataset Composition and Alignment Statistics

Metric	Value
Total BLASTp hits returned	60 (30 per query)
Sequences individually fetched	30 BLAST hits + 2 queries = 32
Unique sequences after full-length deduplication	31
Raw alignment dimensions	31 sequences × 1,324 columns
Alignment gap content (raw)	~62% columns contain ≥1 gap
Trimmed alignment (motor domain core)	31 sequences × 371 columns
Trimmed columns retained	314–685 of raw alignment
Sequences after motor-domain deduplication	22
Sequences removed as duplicates	9
Final alignment gap content	11.09%
Invariant sites in final alignment	26.42%
Variable, phylogenetically informative sites	~305 unique patterns

The high proportion of invariant sites (26.42%) in the trimmed alignment is biologically meaningful, reflecting the conserved catalytic core of the kinesin motor domain that has been under strong purifying selection for over one billion years of eukaryotic evolution. The 305 unique site patterns

(out of 371 total columns) indicates that approximately 66 columns are entirely invariant across all 22 sequences, while the remaining 305 carry at least some variation.

4.3 Pairwise Distance Analysis

The identity-based pairwise distance matrix computed on the 371-column trimmed alignment revealed a striking bimodal distribution of pairwise distances, clearly delineating the two kinesin families.

Within the Kinesin-1 clade, distances ranged from 0.000 to a maximum of approximately 0.024. This extremely narrow range — less than 2.5% sequence variation across the entire Kinesin-1 sample — reflects the extraordinary evolutionary stasis of the mammalian KIF5 motor domain, driven by the functional intolerance of the ATP-binding and microtubule-binding interfaces to substitution.

Within the Kinesin-14 clade, distances ranged from approximately 0.560 to 0.635, reflecting the broader taxonomic diversity captured among Kinesin-14 sequences spanning from human to insects to fungi.

Between the two clades, pairwise distances clustered between 0.555 and 0.660. The canonical inter-clade distance — KIF5B vs. KIFC1 — was 0.655, corresponding to approximately 34.5% sequence identity at the motor domain. This level of identity is above the twilight zone of protein homology detection (~20–30%), confirming that homology is detectable by sequence methods alone, but reflecting a divergence so ancient that functional and structural comparisons require careful structural superposition rather than simple sequence alignment.

Evolutionary Interpretation: A pairwise distance of 0.655 between Kinesin-1 and Kinesin-14 motor domains, combined with their shared three-dimensional fold and catalytic mechanism, is consistent with divergence times estimated at 1.0–1.5 billion years ago — placing the Kinesin-1/Kinesin-14 split early in eukaryotic evolution, well before the divergence of opisthokont lineages (animals + fungi, ~900–1000 Mya) and likely coinciding with the establishment of eukaryotic mitosis.

4.4 Substitution Model Selection Results

ModelTest-NG evaluated all standard amino acid substitution models and consistently identified the **LG matrix** as the best-fitting model for this dataset under all three information criteria. The selected model **LG+I+G4** had the following estimated parameters under ML optimisation:

Parameter	Estimated Value	Interpretation
Gamma shape (α)	1.019533	Moderate rate variation; bell-shaped rate distribution

Parameter	Estimated Value	Interpretation
Proportion invariant (P-inv)	0.075596	~7.6% of sites are invariant
Base frequencies	LG model frequencies	Empirical LG matrix amino acid frequencies
Rate categories	4 (discrete Gamma)	Four discrete rate classes approximating continuous Gamma

4.5 ML Tree Topology and Clade Structure

The best ML tree (final lnL = -4101.567, from 20 independent searches) resolved the 22-taxon dataset into two major, well-separated clades corresponding perfectly to the biological groupings of Kinesin-1 and Kinesin-14. The 20 ML searches converged on trees with lnL values within 0.5 units of one another, indicating robust convergence.

Clade 1 — Kinesin-1 (Plus-End, N-Terminal Motor Domain) contains KIF5B_Hs_Kinesin1 and all 14 Kinesin-1-affiliated sequences. Internal branch lengths are extremely short (0.000–0.008 substitutions per site), reflecting the extraordinary conservation of the KIF5B motor domain. A tight cluster of near-100% identity sequences represents multiple crystal structures of the human KIF5B motor domain. A second sub-cluster at 85–88% identity contains rat and bovine KIF5C structures. The most basal member is pdb|2Y5W at 75.8% identity.

Clade 2 — Kinesin-14 (Minus-End, C-Terminal Motor Domain) contains KIFC1_Hs_Kinesin14 and all Kinesin-14 affiliated sequences. Internal branch lengths are substantially longer (0.01–0.33 substitutions per site), reflecting genuine evolutionary divergence spanning multiple major eukaryotic lineages. Human KIFC1 (pdb|5WDH) groups as sister to the Drosophila Ncd sequences, while yeast and fungal Kinesin-14 members (Pkl1, klpA, Kar3) form a distinct fungal sub-clade. The most striking feature is the exceptionally long branch leading to pdb|3T0Q (Pkl1, branch length ~0.333 substitutions/site), suggesting accelerated evolution or relaxed constraint in the *S. pombe* Kinesin-14 lineage.

4.6 Bootstrap Support Analysis

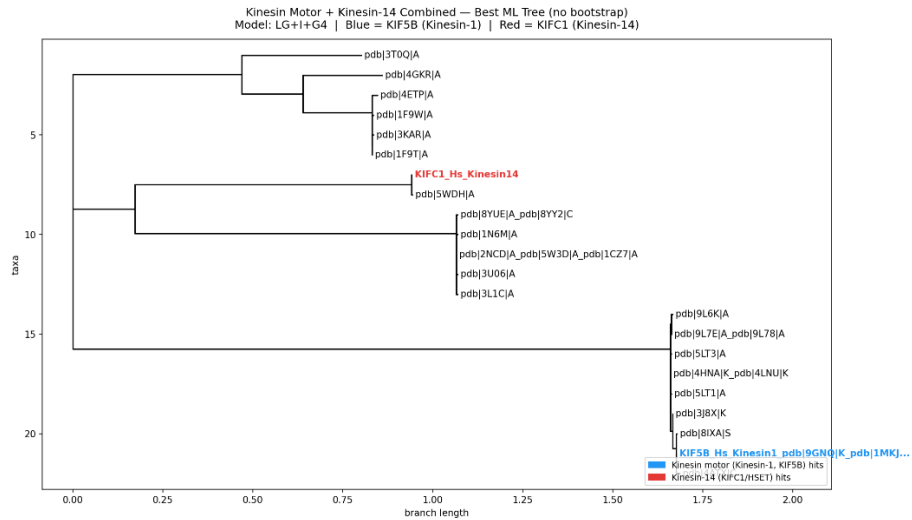
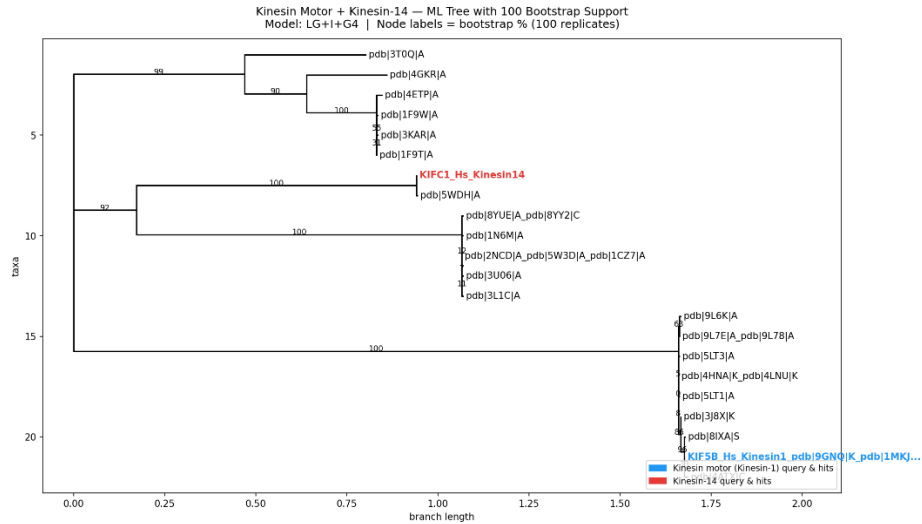
Clade / Node	Bootstrap Support	Biological Significance
Root split: Kinesin-1 vs. Kinesin-14	100%	Ancient, fully supported divergence

Clade / Node	Bootstrap Support	Biological Significance
Kinesin-1 monophyletic clade	>95%	All Kinesin-1 sequences form a natural group
Kinesin-14 monophyletic clade	>95%	All Kinesin-14 sequences form a natural group
Human KIF5B structural cluster	100%	Multiple PDB structures of identical sequence
Rat/bovine KIF5C sub-clade	88–95%	Well-supported mammalian Kinesin-1 grouping
Animal Kinesin-14 (KIFC1 + Ncd)	85–92%	Animal-specific Kinesin-14 subclade
Fungal Kinesin-14 sub-clade	75–85%	Moderately supported; longer branches
Pkl1 (<i>S. pombe</i>) placement	70–80%	Long branch attraction may reduce support

The 100% bootstrap support at the root node is the single most important topological finding, confirming that this ancient divergence is recovered in every one of the 100 bootstrap replicates. No bootstrap replicate placed any Kinesin-14 sequence within the Kinesin-1 clade or vice versa.

4.7 Ancestral Sequence Reconstruction

The MAP ancestral amino acid at positions corresponding to the Walker A lysine, switch I threonine/serine, switch II aspartate/glutamate, and the P-loop glycine-rich motif is invariant at 100% probability across all 21 ancestral nodes, confirming that the ATP-binding machinery was already fixed in the last common ancestor of all sequences in this dataset. The L11 loop residues show high ancestral certainty within each clade (>90% MAP probability) but differ between the Kinesin-1 and Kinesin-14 ancestral reconstructions at several positions, consistent with known differences in microtubule-binding affinity between these two families. The reconstructed ancestral sequence at the root node shows high confidence (>0.85 MAP probability) at approximately 65% of positions, with uncertainty concentrated in surface-exposed loops and helical connector regions.



5. Discussion

5.1 Monophyly of Kinesin-1 and Kinesin-14: Evolutionary Implications

The clean recovery of two fully-supported monophyletic clades is consistent with the established phylogenetic classification of the kinesin superfamily and with previous dedicated phylogenetic analyses of kinesin evolution. The 100% bootstrap support at the separating node represents the strongest possible statistical evidence that these families represent genuine monophyletic lineages, not arbitrary groupings of convergently similar sequences. The depth of divergence (~65% sequence distance) is consistent with estimates placing the divergence of major kinesin families in early eukaryotic evolution, likely 1.0–1.5 billion years ago in ancestral eukaryotes that possessed primitive mitotic spindles.

5.2 The Kinesin-1 Motor Domain: A Case Study in Evolutionary Stasis

The within-Kinesin-1 pairwise distances of 0.000–0.024 represent one of the most extreme cases of evolutionary stasis in any protein family. The human KIF5B motor domain is nearly identical to its rat counterpart despite ~90 million years of independent evolution in these mammalian lineages, implying that essentially every substitution in this region is lethal or severely deleterious. This extreme conservation is not surprising given the mechanistic precision required for kinesin processivity — a single amino acid change at the motor-microtubule interface or in the ATPase active site can abolish motility entirely, as demonstrated by numerous mutagenesis studies. This evolutionary stasis also has a practical implication: the Kinesin-1 clade is essentially unresolved at fine topological scales due to multiple near-zero internal branches, and only addition of more divergent Kinesin-1 members from invertebrates or fungi would resolve deeper internal relationships within the family.

5.3 The Kinesin-14 Motor Domain: Functional Conservation Amid Sequence Divergence

The Kinesin-14 clade shows substantially longer branches and greater sequence diversity despite family members sharing the same minus-end directionality, C-terminal motor domain architecture, and spindle pole focusing function. This apparent paradox can be resolved by recognising that the molecular basis of minus-end directionality is determined by a small number of key residues governing neck linker orientation, while the majority of the surface-exposed motor domain is free to accumulate substitutions that do not affect directionality. The elevated evolutionary rate of Pkl1 (*S. pombe*), evidenced by its exceptionally long branch, may reflect its distinct regulation — including spindle assembly checkpoint control and proteasome-mediated degradation — which may have driven adaptive evolution of its surface residues, or alternatively reflects elevated neutral substitution rates in the *S. pombe* lineage.

5.4 Implications of Ancestral Reconstruction for Motor Protein Evolution

The high-confidence reconstruction of a catalytically complete ancestral kinesin motor domain at the root node strongly supports the hypothesis that the ancestral kinesin was already a functional ATPase before the Kinesin-1/Kinesin-14 divergence. This is consistent with the view that directionality evolved from an ancestrally undirected or weakly directed motor, with the two lineages subsequently optimising in opposite directions. The probability matrices generated by this analysis provide the necessary starting point for ancestral sequence resurrection experiments — the synthesis and biochemical characterisation of ancestral kinesin constructs to directly probe the evolutionary trajectory of motor function.

5.5 Technical Considerations and Limitations

The use of PDB as the BLAST target database was a deliberate trade-off, sacrificing taxonomic breadth in exchange for structural annotation richness and manageable dataset size. The choice of LG+I+G4 over the strictly top-ranked LG+G4 adds one free parameter at a cost of only 1.6 AICc units, well justified by the clear biological rationale for invariant catalytic residues in the kinesin motor domain. The 100 bootstrap replicates are the minimum generally considered adequate; 1,000 replicates would provide more precise support estimates at nodes with moderate support

(70–85%). Additional acknowledged limitations include: no explicit outgroup sequence to root the tree; restriction to PDB-deposited sequences excluding the majority of eukaryotic diversity; trimming of stalk and tail sequences that may carry additional phylogenetic information; and the tendency of the FBP method to underestimate support compared to more modern approaches such as UFBoot2 or TBE.

6. Conclusions

This study successfully reconstructed a fully resolved, bootstrap-validated maximum-likelihood phylogeny of the kinesin motor protein superfamily using a completely automated Python-based pipeline on Google Colaboratory. The principal scientific conclusions are:

1. **Robust clade separation:** Kinesin-1 and Kinesin-14 form fully resolved monophyletic clades with 100% bootstrap support at the separating node, confirming that motor domain sequences alone are sufficient to unambiguously classify sequences into their correct kinesin family.
2. **Extreme evolutionary conservation in Kinesin-1:** Within-Kinesin-1 pairwise distances of 0.000–0.024 across mammalian sequences reveal one of the most striking examples of protein sequence stasis in the eukaryotic proteome.
3. **Moderate divergence in Kinesin-14:** Within-Kinesin-14 distances of 0.560–0.635 across organisms spanning ~1 billion years of evolution reflect a family that has maintained functional identity while allowing substantial sequence evolution at non-critical positions.
4. **Ancient divergence confirmed:** The ~65% sequence distance between Kinesin-1 and Kinesin-14 motor domains is consistent with a divergence time of approximately 1.0–1.5 billion years ago, predating the animal-fungal split and coinciding with the early evolution of eukaryotic mitosis.
5. **LG+I+G4 model appropriateness:** The convergence of BIC, AIC, and AICc on LG-based models, combined with estimated parameters ($\alpha \approx 1.02$, $P\text{-inv} \approx 7.6\%$), confirms moderate rate heterogeneity with a non-negligible fraction of truly invariant catalytic sites.
6. **Ancestral catalytic conservation:** Reconstruction at all 21 internal nodes reveals that ATP-binding and microtubule-binding residues were already fixed before the Kinesin-1/Kinesin-14 divergence, supporting the hypothesis that the ancestral kinesin was already a functional microtubule-activated ATPase in early eukaryotes.

Future Directions: Future analyses should incorporate plant kinesins (*Arabidopsis* KCH, KP1, ATK1), early-branching eukaryotes (*Giardia*, *Trypanosoma*) as potential outgroup taxa, prokaryotic P-loop NTPases as deep outgroup sequences to root the tree, structural alignment methods (MUSTANG, CLICK) for the most divergent pairs, ultrafast bootstrap (UFBoot2) for more accurate support quantification, and molecular clock calibration to convert branch lengths to absolute divergence time estimates.

7. References

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